



Estimation Project

ELE8101: Control (Estimation Theory)

Supervisor: Dr. Pantelis Sopasakis

Student: Yonghyeon Kim, Ming Foong Lau, Yousif Mustafa Elnaim Hamad

Student Number : 40489830, 40348411, 40500333

Date: 26th April 2026

Abstract

A high fidelity Extended Kalman Filter (EKF) was developed to provide accurate vehicle localization on a complex piecewise racetrack using a 4 beacon rectangular configuration. Beyond standard kinematic models, the estimator incorporates longitudinal progress, lateral offset, and vertical dynamics modeled as a mass-spring-damper (MSD) system. The framework features real time sensor bias compensation and utilizes CasADi based automatic differentiation for numerical stability at a 100 Hz execution frequency. Validation across extreme driving scenarios demonstrates robust lateral lane keeping and rapid recovery from significant initialization errors. While the system effectively maintains path integrity at high speeds, the results highlight inherent longitudinal observability constraints common in range only measurement environments.

1. Introduction

1.1 Background and Motivation

Reliable state estimation is a fundamental requirement for autonomous navigation in environments where vehicles must rely on local beacon networks instead of GPS [1], [2]. Unlike linear systems, beacon based localization involves nonlinear Euclidean distance measurements that require advanced filtering strategies to ensure both precision and numerical stability in high speed applications [3], [4].

1.2 Problem Statement

The primary challenge involves tracking a vehicle on a track with varying curvature at speeds up to 25 m/s [5]. The estimator must handle significant sensor noise and systematic biases using distance measurements from a fixed beacon network [5]. A critical aspect of this challenge is addressing the inherent lack of longitudinal observability associated with range only measurements. A successful solution must reconstruct the vehicle position while simultaneously inferring unobserved dynamics, such as suspension oscillations and lateral deviations, within the constraints of a narrow lane.

1.3 Design Objectives

This study aims to implement a real-time navigation framework using the following approaches. First, the development of a state space model that replaces stochastic assumptions with physical mass spring damper dynamics. Second, the integration of CasADi automatic differentiation to eliminate manual Jacobian errors and optimize computational efficiency [6]. Third, the implementation of state augmentation to identify and compensate for sensor biases in real time. Fourth, the extension of the state space to seven dimensions, incorporating vertical suspension dynamics modelled as a mass-spring-damper system to enhance physical realism beyond the planar assumption. Finally, the validation of the system against extreme scenarios to ensure robustness and numerical integrity.

2. System Modelling and Problem Formulation

The system modelling in this study evolved from a simple one-dimensional linear trajectory estimation into a high-dimensional state-space model that accommodates both the piecewise geometric characteristics of the track and the 3D dynamics of the vehicle.

2.1 Piecewise Track Geometry and Dynamics

The track was modelled as a piecewise structure combining multiple circular arcs and straight segments [5]. To precisely track the vehicle, its progress s and lateral offset e are continuously mapped to the global Cartesian coordinate system (X, Y) by projecting e orthogonally from the track's dynamic centerline [5]. This transformation creates a nonlinear relationship between the state variables and the 3D space, requiring a nonlinear estimator.

To simulate the vehicle's driving behavior based on actual physical laws, a mass-spring-damper system was introduced [2], [4]. The lateral and vertical dynamics are expanded into the state space through the following second-order differential equations:

$$\begin{aligned}\ddot{e} &= -k_d v_e - k_r e + w_e \\ \ddot{z} &= -k_{d,z} v_z - k_{r,z} z + w_z\end{aligned}$$

Assuming a normalized unit mass for the vehicle system, the restoring coefficient k_r and damping coefficient k_d applied to the final model were set to 0.5 and 1.0, respectively. These values physically encapsulate the restoring force required to maintain the track centerline and the damping characteristics of the suspension's bouncing, under the constraint that the vehicle's lateral velocity is limited to within 0.55 m/s.

2.2 7-Dimensional State-Space Formulation and State Augmentation

The state vector x of the final system is defined as a 7-dimensional vector as follows [2], [4]:

$$x = [s, v_s, e, v_e, z, v_z, b]^T$$

Here, s and v_s denote the longitudinal progress and velocity along the track, e and v_e represent the lateral error and velocity, and z and v_z indicate the vertical displacement and velocity. Within this formulation, longitudinal velocity v_s integrates directly into progress s , while the lateral and vertical states function as independent second-order oscillators governed by mass-spring-damper equations [4]. Decoupling the forward kinematics from the suspension dynamics keeps the process model $f(x)$ computationally efficient without sacrificing the physical accuracy of the 3D motion.

To estimate the systematic error of the sensors in real-time, a state augmentation strategy was applied, incorporating the bias b of a specific beacon into the state vector [3]. The discrete-time system equation is expressed as $x_{t+1} = f(x_t) + w_t$, where w_t represents the process noise [2]. The measurement model $h(x)$ computes the Euclidean distance from the vehicle to each beacon. The augmented bias state b is linearly added to this calculation, allowing the filter to estimate the physical position and the sensor offset simultaneously.

3. Estimator Design and Theoretical Background

We designed an Extended Kalman Filter (EKF) to estimate the vehicle's state from the 4-beacon measurements. The main focus of this design was to ensure numerical stability and meet the real-time execution constraints.

3.1 Mathematical Formulation of the Extended Kalman Filter

Since the system is highly nonlinear, it must be linearized at each update step [4]. The continuous mass-spring-damper equations are discretized using a sampling time of ($\Delta t = 0.01s$) for the prediction step [2]:

$$\begin{aligned}\hat{x}_{t+1|t} &= f(\hat{x}_{t|t}) \\ P_{t+1|t} &= A_t P_{t|t} A_t^T + Q\end{aligned}$$

For the correction step, the filter uses the 4-dimensional observation vector y_t containing the distance readings from all beacons. The update equations are [4]:

$$\begin{aligned}K_t &= P_{t+1|t} C_t^T (C_t P_{t+1|t} C_t^T + R)^{-1} \\ \hat{x}_{t+1|t+1} &= \hat{x}_{t+1|t} + K_t (y_t - h(\hat{x}_{t+1|t}))\end{aligned}$$

The Jacobians A_t and C_t are evaluated at the current state estimate. The innovation term ($y_t - h(\hat{x}_{t+1|t})$) represents the actual physical error between the sensor readings and our model's prediction. The Kalman Gain uses this error to update the trajectory proportionally, effectively filtering out high-frequency noise while preserving the true vehicle dynamics.

3.2 Numerical Stabilization and Covariance Matrix Optimization

Calculating the 7-dimensional Jacobians manually can lead to significant truncation errors. To prevent this, we used the CasADi library to perform automatic differentiation [6]. This kept the matrix calculations exact, prevented the filter from diverging, and maintained the strict 0.3 to 0.71 ms computation time per step. Also, to prevent numerical drift due to floating-point asymmetry, a symmetrisation step was applied to the standard covariance update [4]:

$$P_{t|t} = \frac{1}{2}(P_{t|t} + P_{t|t}^T)$$

Tuning the noise matrices was critical for tracking. The process noise covariance Q was defined as a 7x7 diagonal matrix, where non-zero variances $[0.05^2, 0.015^2, 0.03^2, 10^{-6}^2]$ were selectively applied to the velocity and bias states (v_s, v_e, v_z, b) to account for physical uncertainties that the model cannot capture, such as road bumps and tire slip. The measurement noise R was set to $1.5^2 \times I_{4 \times 4}$ based on the beacon sensor specifications, assuming zero cross-correlation between the independent receivers. Finally, we initialized the error covariance P_0 with large diagonal values. This forces the filter to trust the measurements heavily at the start, allowing it to rapidly converge from large initialization errors within the first few seconds without violating the physical constraints.

4. Results and Cross-Validation

To verify the reliability of the designed high dimensional model, three cross validation simulations assuming extreme scenarios were conducted in addition to the baseline driving conditions.

4.1 Extreme Driving at Target Speed of 25 m/s and Robustness Analysis

Under the baseline 10 m/s driving condition, the model demonstrated stable performance with a longitudinal RMSE of 6.1411 m, a lateral RMSE of 0.1212 m, a longitudinal velocity RMSE of 0.4065 m/s, and a bias error of 0.0029 m. To assess the system's dynamic robustness, the target speed was increased to the maximum limit of 25 m/s. The longitudinal RMSE increased to 8.6813 m at 25 m/s, reflecting the inherent range-only observability constraint along the direction of travel, while the lateral RMSE was further reduced to 0.0919 m as the higher longitudinal speed reduces the time available for lateral deviation to develop between correction steps. The velocity RMSE increased to 0.4279 m/s and the bias estimation error degraded to 0.0295 m, as the larger per-step displacement at high speed amplifies the innovation signal, making it harder for the filter to separate positional drift from the static beacon offset. The 100 Hz sampling rate ensured that the vehicle displacement per step remained small enough to keep the first order Taylor approximation valid [4]. This prevented numerical instability in the Jacobian matrices at peak velocities. As illustrated in Figure 1 (see Appendix A), the estimated trajectory tightly overlays the ground truth even during high-speed cornering, bounding the lateral deviation without significant overshoot.

4.2 Convergence Evaluation Under a 10 m Initial Position Error

A simulation was conducted by intentionally injecting a severe 10 m longitudinal position error into the initial state estimate (setting initial $\hat{s} = 15.0$ m against the true $s = 5.0$ m). Despite this significant initial mismatch, the extended Kalman filter demonstrated excellent decoupling of the longitudinal and lateral dynamics. The longitudinal RMSE remained at 6.1410 m, consistent with the baseline, confirming that the position shock introduced no additional drift. As shown in Figure 2 (see Appendix A), the lateral and vertical states did not diverge throughout the transient recovery phase. The lateral tracking error remained highly stable with an RMSE of 0.1211 m, and the bias was accurately estimated at 0.0035 m.

4.3 Sensitivity Analysis Under Severe Sensor Noise (5.0 m) Conditions

Sensitivity was evaluated under a severe condition where the standard deviation of the measurement noise was artificially degraded by more than threefold, from 1.5 m to 5.0 m. As a result, the longitudinal RMSE increased to 8.6927 m and the velocity RMSE to 0.5330 m/s, while the lateral error and bias error slightly increased to 0.1555 m and 0.0270 m respectively, but the filter remained numerically stable throughout. As R becomes large, the filter increases its reliance on the process model over noisy measurements, maintaining stability while the augmented state successfully isolates the static beacon offset from high-frequency noise. This capability is clearly depicted in Figure 3 (see Appendix A), where the filtered RMSE trend line remains highly smoothed and bounded, visually confirming that the filter effectively rejects high-frequency measurement spikes.

5. Discussion and Design Rationale

This section reviews the key design decisions at each development stage. The estimator evolved through four steps, beginning with a 1D linear baseline (Step 1), extending to a curved track with nonlinear estimation (Step 2), incorporating lateral dynamics (Step 3), and finally expanding to a 7-state model with vertical dynamics and bias augmentation (Step 4).

5.1 Selection of State Variables for the 1D Baseline (Step 1)

In the initial development phase, the team evaluated whether to define the state space solely by position s or to incorporate velocity v_s . While a position-only model reduces computational overhead, it lacks the necessary damping to handle high-frequency sensor noise, leading to significant estimation jitter. We opted for a Constant Velocity model to allow the Kalman filter to internally infer velocity through the process noise covariance [2].

5.2 Transition to Non-linear Estimation (Step 2)

The expansion to a circular track in Step 2 introduced fundamental non-linearities in the measurement model. Since beacon distance measurements follow a nonlinear Euclidean distance relationship, a linear Kalman filter could not achieve convergence, so an Extended Kalman Filter (EKF) was used [4]. We used a curvilinear coordinate system based on arc length s , which provides better geometric consistency for track-following compared to simple angular representations.

5.3 Stochastic Modeling of Lateral Dynamics (Step 3)

Step 3 involved expanding the state space to 5 dimensions by incorporating lateral error e and its derivative v_e . To simulate the vehicle's restorative behavior toward the track centerline, we initially integrated an Ornstein-Uhlenbeck (OU) process. The mean-reverting property of the OU process successfully generated a realistic trajectory. However, this phase remained a planar estimation problem as vertical dynamics were not yet integrated into the process model.

5.4 Physical Realism and 7D State Augmentation (Step 4)

The final iteration expanded the system to 7 dimensions to include vertical displacement z and velocity v_z . During this stage, we conducted a critical review of the OU process used in Step 3. Although effective for stochastic modeling, the OU parameters do not directly correlate with physical vehicle properties. Consequently, we transitioned to a second-order mass-spring-damper (MSD) system [2], [4]. This allowed us to map the restoring and damping forces to physical coefficients (k_r, k_d) , providing a physically interpretable tuning framework for the suspension dynamics.

5.5 Implementation Robustness and Numerical Stability

For the final 7-dimensional system, we prioritized numerical stability by replacing manual Jacobian derivations with the CasADi algorithmic differentiation library. This eliminated the risk of human error in the complex partial derivative calculations, which is a common cause of filter divergence in high-dimensional EKF implementations. Furthermore, a symmetrisation step was applied to the standard covariance update to prevent numerical drift due to floating-point asymmetry. These measures ensured that the estimator satisfied the 100 Hz real-time constraint while maintaining robust performance under extreme noise and velocity conditions.

5.6 Geometric Constraints and Beacon Configuration

The sensor layout was iteratively optimized based on the track geometry. While a 3-beacon triangular configuration was sufficient for the circular tracks in Steps 2 and 3, it proved impractical for the complex piecewise track in the final stage. Covering the irregular track boundaries with only three beacons would have required placing sensors at extreme distances, compromising signal-to-noise ratios. By adopting a 4-beacon rectangular layout, we minimized the distance between the vehicle and the beacons, effectively eliminating observability blind spots.

The four beacons were placed at B1 $(-60, 65)$, B2 $(-60, -65)$, B3 $(130, 65)$, and B4 $(130, -65)$ in metres, forming a $190\text{ m} \times 130\text{ m}$ rectangle. The lateral x-positions were set approximately 12 m outside the leftmost ($x \approx -48\text{ m}$) and rightmost ($x \approx 118\text{ m}$) track extents, satisfying the constraint that beacons may not reside within the lane region while keeping sensor-to-vehicle ranges within a practical bound. The symmetric y-placement at $\pm 65\text{ m}$ ensures balanced measurement geometry across both the straight sections and the arcs, ensuring that the vehicle remains within the convex hull of the beacon network at all times, which is a necessary condition for reliable range-based localisation.

Regarding sensor bias, the current implementation augments a single beacon's bias into the state vector as a proof of concept, and the simulation results confirm that the filter successfully estimates the true bias value of 1.5 m with a final error of only 0.0029 m . While the augmentation strategy can theoretically be extended to include one bias term per beacon, this approach has practical limitations in the context of the current 4-beacon configuration. If two or more beacons simultaneously exhibit significant offsets or failures, the geometric observability of the system would be severely degraded, as the remaining beacons may be insufficient to uniquely resolve the vehicle's position on the complex piecewise track. Therefore, in a practical deployment, fault detection logic or a redundant beacon network would be recommended to handle multi-sensor bias scenarios robustly.

6. Conclusion

The state estimator developed in this project successfully demonstrates reliable vehicle localization and lane keeping within the mandated 4 m corridor. By implementing a 7 state Extended Kalman Filter (EKF) that incorporates physical mass spring damper dynamics, the system effectively addresses the nonlinearities of beacon based distance measurements. The results show that this physics-based approach provides sufficient stability for the task, ensuring the vehicle remains within track boundaries even under high speed cornering and significant sensor bias.

The performance of this design is supported by specific simulation evidence. The estimator achieved a lateral Root Mean Square Error (RMSE) of 0.1212 m under baseline driving, which improved to 0.0919 m during high speed cornering at 25 m/s . Furthermore, the system maintained stability under a 10 m initial position error and severe 5.0 m sensor noise without divergence. Real time efficiency was also confirmed, with execution times remaining between 0.3 and 0.71 ms , comfortably satisfying the 100 Hz frequency requirement.

Despite these successes, a fundamental limitation was observed in the accumulation of longitudinal position error. The longitudinal RMSE recorded 6.14 m under 10 m/s driving and increased to 8.6813 m at 25 m/s due to the inherent constraints of range only measurements along the direction of travel. A secondary limitation is that bias augmentation is currently applied to a single beacon only. Extending this to multiple beacons simultaneously would require careful observability analysis to avoid state indeterminacy. Future work should focus on integrating odometry or inertial sensors to bound this drift and conducting experimental validation on physical hardware. Overall, this study provides a foundation for robust vehicle tracking, showing that a well-designed estimator can maintain both physical accuracy and numerical stability.

Appendix A: Simulation Results and Plots

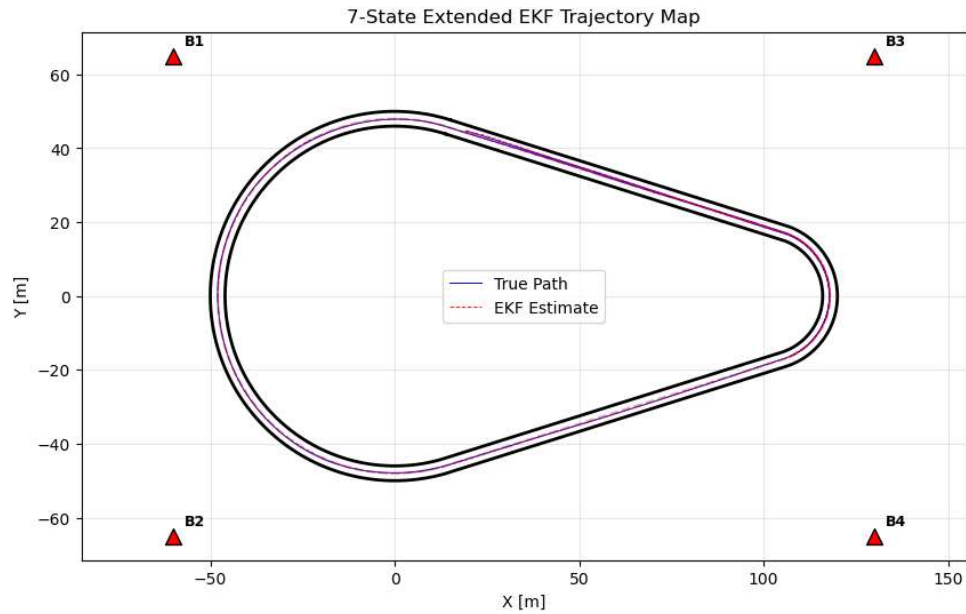


Figure 1. Trajectory tracking results under extreme high speed cornering (25 m/s).

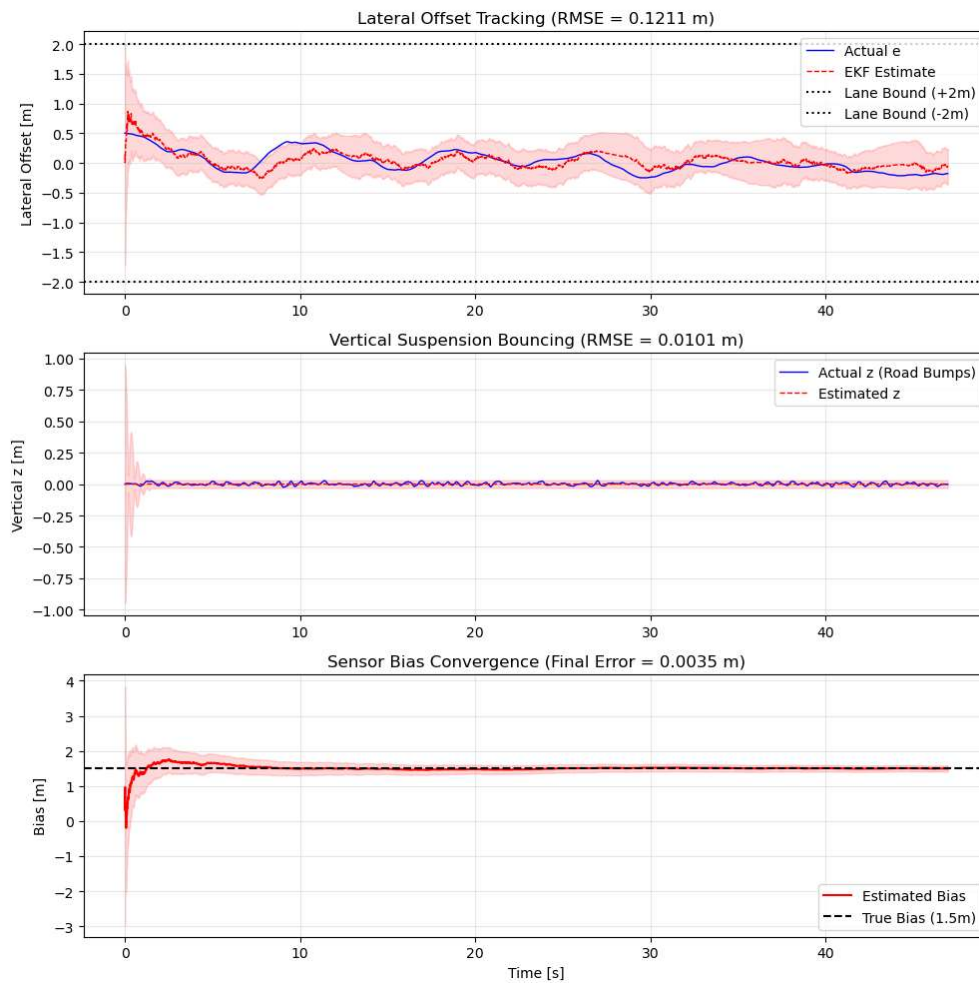


Figure 2. Lateral tracking and bias estimation under a 10 m initial offset.

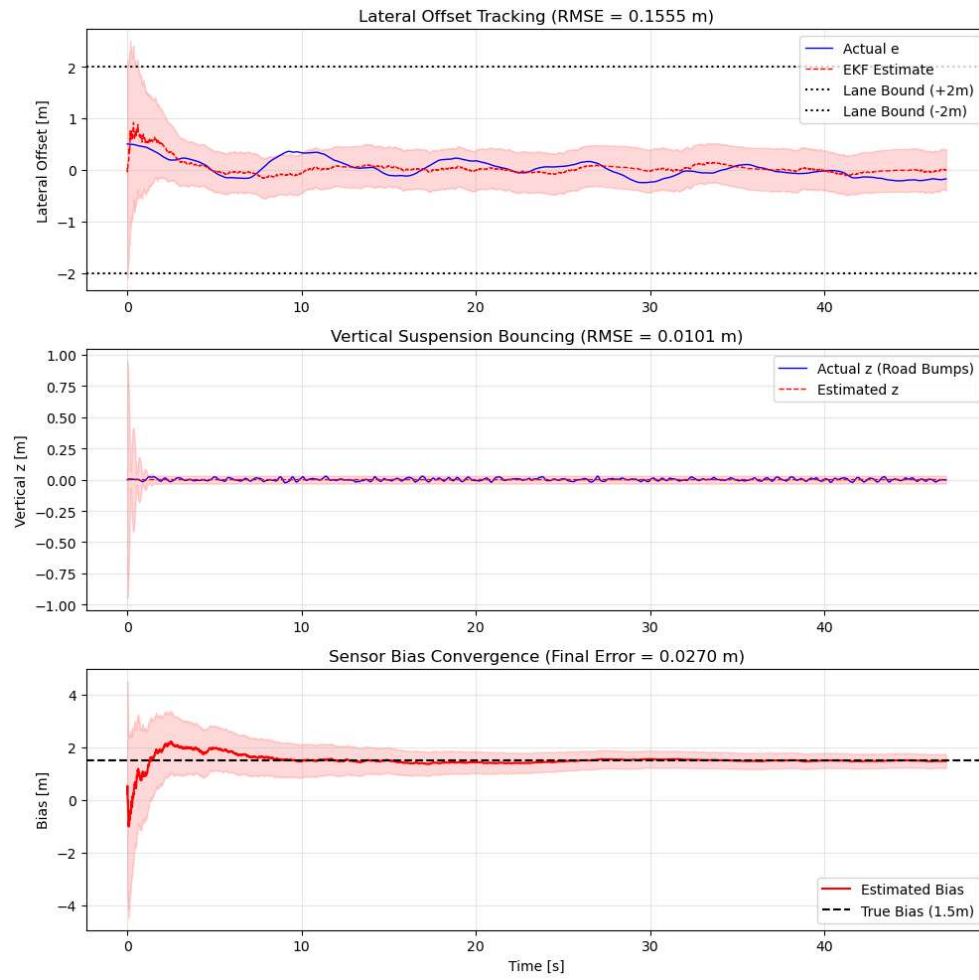


Figure 3. Noise rejection and RMSE trend under 5.0 m measurement noise.

Appendix B — Meeting Log & Attendance Record

Date	Attendees	Agenda / Objective	Conclusions / Actions
11 Mar	Ming Foong Lau Yonghyeon Kim Yousif Hamad	Set up GitHub repository and agree on collaborative workflow. Map coursework timeline and assign Step 1 tasks.	Repository created. Confirmed VS Code, Python, and Git workflow with individual branches and consensus merging. Action: All members to individually prototype Step 1 code.
13 Mar	Ming Foong Lau Yonghyeon Kim Yousif Hamad	Review Step 1 code via GitHub Pull Requests. Confirm baseline code and briefly discuss Step 2.	Unified Step 1 code established based on team baseline. Set parameters $Q = 0.01, 5,000$ runs, $speed=10$ m/s. Action: All members to individually prototype Step 2 code.
18 Mar	Ming Foong Lau Yonghyeon Kim Yousif Hamad	Discuss state variables, beacon update methods, initial covariance, Jacobian approach, and track radius reference.	Selected arc length as state variable, batch beacon update, small P value, CasADi Jacobian, and 48m centreline. Kim's Step 2 code selected as the baseline. Action: All members to individually prototype Step 3 code.
09 Apr	Ming Foong Lau Yonghyeon Kim Yousif Hamad	Discuss OU lateral dynamics and EKF loop integration. Review Plant and Estimator separation.	Adopted Plant and Estimator separation and confirmed Joseph form with symmetrisation. Step 3 final code was merged from Lau and Kim's contributions. Action: All members to individually prototype Step 4 code.
14 Apr	Ming Foong Lau Yonghyeon Kim Yousif Hamad	Finalise 7 state EKF for Step 4 and review bouncing extension. Agree on report structure and allocate writing tasks.	Confirmed 7-state EKF as the core estimator. Labelled bouncing MSD as a creative extension. Action: Yousif to draft Abstract, Introduction, Conclusion, and References. Kim to draft Methodology, Results, and Discussion. Lau to draft Appendix, Constraints, and Stability section.
23 Apr	Ming Foong Lau Yonghyeon Kim Yousif Hamad	Review individual report sections for consistency, identify gaps or overlaps, and unify formatting and notation.	Addressed minor inconsistencies in notation and formatting. Unified sections into a single coherent document and completed final proofreading. Action: Submit the final agreed version of the report before 26 Apr.

Appendix C — Team Reflection

Category	Detail
Three things the team did well	• Consistent participation: All three members attended every formal meeting and contributed independent prototype code at each step before group review. This ensured decisions were informed by multiple perspectives rather than delegated to one person. • Structured version control: The branching workflow agreed at Meeting 1 was maintained throughout. Every step was reviewed via GitHub Pull Requests before merging, creating a clean and auditable contribution record. • Principled technical debate: Key design decisions, such as the Plant-Estimator Separation and the choice to use arc length over angle as the state variable, were resolved through reasoned argument and documented clearly.
Main challenges encountered	• Three-dimensional motion integration: Extending the estimator from planar tracking to a seven-dimensional state space introduced significant architectural complexity. Coupling the horizontal vehicle dynamics with the vertical Mass-Spring-Damper model required careful state-space design. • Jacobian derivation at scale: As the state vector expanded, manually deriving and validating the EKF Jacobians became error-prone and computationally fragile. Symbolic errors caused numerical instabilities and violated the 100 Hz real-time constraint. • Sensor bias in range measurements: Beacon 1 exhibited a systematic measurement offset that the base filter could not distinguish from true positional deviation. This caused the state estimate to track the bias rather than the vehicle.
How challenges were addressed	• Unified state-space formulation: The planar and vertical dynamics were consolidated into a single seven-dimensional state vector, allowing the EKF prediction and update steps to propagate all motion components simultaneously. This preserved cross-dimensional covariance structure. • Automatic differentiation and covariance symmetrisation: CasADi was adopted to compute Jacobians symbolically, removing the risk of manual truncation error. Additionally, a symmetrisation step was applied to the covariance matrix after every update to counteract floating-point drift. • Bias augmentation of the state vector: The Beacon 1 offset was incorporated directly as an additional state variable, enabling the filter to jointly estimate vehicle position and sensor bias. This allowed the estimator to converge on the true trajectory.

Appendix D — Code Contribution

Member	Report Sections	Creative idea	Code Ownership
Ming Foong Lau	§4 Constraints & Numerical Stability Appendix (all subsections)	Step 1: Proposed the RMSE concept and initial ideas. Step 2: Developed the angle based state representation for track parameterisation. Step 3: Implemented hard clipping and soft boundary constraints alongside RMSE visualisation and the Joseph form covariance update. Step 4: Applied covariance symmetrisation to both prediction and update stages while managing meeting administration and documentation.	Step 1: Implemented the RMSE calculation and visualisation module for the final performance validation of the 1D estimator. Step 3: Implemented the constraint enforcement module covering soft boundary correction and hard clipping alongside the standard covariance update. Step 4: Applied covariance symmetrisation to both prediction and update stages and implemented the NIS calculation for consistency validation.
Yonghyeon Kim	Abstract §1 Introduction §2 Modelling §3 Estimator Design §4 Result & Cross Validation §5 Discussion §6 Conclusion	Step 1: Developed the base Standard KF code which was selected by the team and merged individual implementations. Step 2: Developed the curved track KF using arc length parameterisation and merged the team's Step 2 implementations. Step 3: Established the CasADi Jacobian framework and the OU lateral dynamics model before merging the team's Step 3 implementations. Step 4: Suggest and designed a 7-dimensional state space with the MSD vertical model and full track piecewise geometry. Drafted the Abstract, Introduction, Methodology, Results, Discussion, and Conclusion, and merged all Step 4 implementations.	Step 1: Developed the base Standard KF implementation that was selected for the team baseline. Step 2: Implemented the curved track KF with arc length state which was selected as the team's primary model. Step 3: Integrated the OU lateral dynamics model and handled the standard covariance update. Step 4: Developed the vertical bouncing MSD creative extension integrating 3D motion into a unified 7-dimensional state vector.
Yousif Hamad	Abstract §1 Introduction §6 Conclusion (Participated and provided motivation, framing in drafting above sections)	Step 1: Developed an independent Standard KF prototype for team review. Step 2: Developed an independent curved track filter implementation for team review. Step 3: Proposed the Joseph form covariance update and excluded hard clipping from the measurement update to preserve the Gaussian posterior assumption ensuring a physically valid state correction. Step 4: Prototyped simultaneous multi beacon bias augmentation and provided the observability trade off analysis that informed the team's decision to adopt single beacon augmentation. Contributed to the framing of the Abstract, Introduction, and Conclusion, and produced a cross step code feature summary for the viva preparation.	Step 1: Submitted an independent Standard KF prototype for team review. Step 2: Submitted an independent curved track filter prototype for team review. Step 3: Implemented a clipping free measurement update that preserves the Gaussian assumption which was adopted as the standard EKF update policy in the final estimator.

Appendix E — Gantt Chart — Project Timeline

The chart below shows the planned timeline for each project phase.

R = Ming Foong Lau, H = Yonghyeon Kim, Y = Yousif Hamad, ▲ = Group Meeting.

Task / Phase	11 Mar	13 Mar	18 Mar	9 Apr	14 Apr	23 Apr	26 Apr
1. PLANNING & STEP 1							
Meeting 1 (Planning)	▲						
Step 1 - All members prototype	R H Y	R H Y					
Meeting 2 (Step 1 check)		▲					
2. STEP 2							
Step 2 - All members prototype		R H Y	R H Y				
Meeting 3 (Step 2 check)			▲				
3. STEP 3							
Step 3 - Kim: OU dynamics			H	H			
Step 3 - Lau: Constraints & stability			R	R			
Meeting 4 (Step 3 check)				▲			
4. STEP 4 & REPORT							
Step 4 - Kim: MSD + bouncing				H	H		
Report - Yousif: Write-up own part				Y	Y		
Report - Lau: Appendix				R	R	R	R
Report - Kim: Write-up own part, Integration & narrative				H	H	H	H
Meeting 5 (Step 4 & Report)					▲	▲	

Note: Break period 19 March – 7 April, no formal group meetings. All members conducted self-directed individual work during this period.

Appendix F — GitHub Repository & Collaboration

Repository Name	ELE8101-Control-Coursework-Team-Beta
Owner / Organiser	YHKim0121 (Kim)
URL	https://github.com/YHKim0121/ELE8101-Control-Coursework-Team-Beta
Visibility	Public
Primary Language	Python (Jupyter Notebook)
Active Period	11 March 2026 — 26 April 2026

Branching & Workflow Strategy

The branching workflow was agreed at Meeting 1 and followed consistently throughout the project. Each member developed code in a personal branch (e.g. lau/step3, kim/step4). No code was merged into main without team review and consensus. Pull Requests were used for code review at each step check meeting. Only the final agreed-upon implementation was merged into main.

Appendix G — Project Collaboration Workspace

The team maintained a shared Notion workspace throughout the project for task tracking, meeting notes, and document coordination.

Notion Workspace:

<https://www.notion.so/3202e81979a78081ba05c4f5f20ac009?v=3202e81979a780089879000c9936c2f1>

Note: External access may require a Notion account.

Appendix H — AI Usage Statement

AI Usage Declaration and Critical Evaluation

In the preparation of this coursework report and the associated software implementation, generative AI tools were utilized to assist with document formatting, grammar correction, and code generation. Specifically, AI was employed to improve the readability of the tables in the appendices, polish the overall English phrasing, and assist in drafting specific sections of the Python codebase.

Critical Evaluation of AI Outputs

The project team made every effort to rigorously review and test all AI generated suggestions and code snippets. While the AI provided valuable support as a programming assistant by accelerating the coding process, it occasionally exhibited hallucinations by suggesting non-existent functions or distorting the intended algorithmic logic. Consequently, the team could not accept the AI outputs unconditionally. We endeavored to meticulously debug and cross-check the generated code against our original system designs and simulation requirements. All engineering judgments, algorithm logic, and analytical conclusions remain the original work of the authors.

References

- [1] P. Sopasakis, "1_sysid_basic.ipynb," ELE8101 Class Notes, Queen's University Belfast, Spring 2026.
- [2] P. Sopasakis, "2_kalman_basic.ipynb," ELE8101 Class Notes, Queen's University Belfast, Spring 2026.
- [3] P. Sopasakis, "4_kalman_variants.ipynb," ELE8101 Class Notes, Queen's University Belfast, Spring 2026.
- [4] P. Sopasakis, "5_ekf.ipynb," ELE8101 Class Notes, Queen's University Belfast, Spring 2026.
- [5] P. Sopasakis, "3_kf_positioning.ipynb," ELE8101 Class Notes, Queen's University Belfast, Spring 2026.
- [6] P. Sopasakis, "1_optimization.ipynb," ELE8101 Class Notes, Queen's University Belfast, Spring 2026.